Dang Minh Duc

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Education

- **VNU University of Engineering and Technology** — Ha Noi, Viet Nam
  Final-year student in Electronics and Telecommunications; GPA: 3.04 — *July 2023 - Dec 2026*

Skills Summary

- **Languages**: Python, C, C++, Bash
- **Frameworks**: Scikit, TensorFlow, Keras
- **Tools**: Docker, GIT, CubeMX, easyEDA, Kicad
- **Platforms**: Linux, Windows, Arduino, Raspberry, STM32, Segger

Experience

- **Viettel High Tech** — Offline
  Embedded Intern — *July 2026 - August 2026*
  - **RTOS**: Learned and worked with FreeRTOS and ZephyrOS on STM32 MCUs; evaluated and compared the two RTOS platforms.
  - **PetaLinux**: Created a PetaLinux project from an existing Vivado design; configured PetaLinux and an RTOS to run on the two cores of a Xilinx SoC, and implemented inter-core communication using RPMsg for data transmission and reception.
- **Department of Telecommunication Systems** — Offline
  Researcher and Project Coordinator — *July 2024 - Present*
  - **Training and Mentoring**: Organized seminars and trained members in embedded systems and IoT technologies.
  - **UWB Indoor Positioning Research**: Conducted research on Ultra-Wideband indoor positioning systems, including system design, algorithm development, and performance analysis.
- **Vietnam Electronic Design Competition 2025** — Offline
  System Developer — *Aug 2025 - Nov 2025*
  - **Competition Theme**: Embedded systems and AIoT technologies for intelligent and connected systems.
  - **Team Project**: Developed an intelligent positioning and navigation system for autonomous mobile robots in factory environments.

Projects

- **UWB-Based Indoor Robot Localization with State Estimation Filters**:
  - Designed and built a complete UWB indoor positioning system using DWM1001 modules for real-world data collection.
  - Implemented and benchmarked EKF, UKF, and PF for nonlinear state estimation; identified UKF as the optimal trade-off between accuracy and computational cost for real-time robotics.
  - Results published at REV-ECIT conference and *Journal on Electronics and Communications*.
  - **Tech:** DWM1001, Python, C embedded, Raspberry Pi 4

- **Smart Parking System** | Course Project:
  - **Role:** System planning, architecture design, and integration testing.
  - Designed the overall system flow and component interaction: defined the Vehicle-to-Slot Association (VSA) pipeline connecting RFID gate control, YOLOv8 + BotSORT camera tracking, IR slot sensing, and MQTT communication layer.
  - Planned integration between hardware nodes (ESP8266, Arduino Nano 33 IoT) and software layers (AI camera, backend, frontend), ensuring consistent data flow across all subsystems.
  - End-to-end testing of violation detection rules (double-parking, excessive roaming, reserved-slot theft) and verified fee calculation logic against defined scenarios.
  - **Tech:** ESP8266, Arduino Nano 33 IoT, YOLOv8, BotSORT, Python, RFID RC522, IR sensor, HiveMQ, MongoDB

- **AMR for Light Cargo Transport**:
  - Built an AMR using Raspberry Pi 5 as central processor, fusing IMU (BNO055) and wheel odometry (GA25-370 encoders) via STM32 Bluepill for reliable pose estimation.
  - Implemented dual-layer cargo verification: Pi Camera V2 scans barcodes to confirm cargo identity; load cell validates expected weight — rejecting mismatched or tampered packages before transport.
  - **Tech:** Raspberry Pi 5, STM32 Bluepill, BNO055, GA25-370 encoder, Pi Camera V2, load cell, Python, STM32 HAL

- **Lightweight UWB NLOS Identification for IoT Edge Devices**:
  - Developed a lightweight UWB NLOS classifier using Channel Impulse Respond (CIR)
  - Achieved 93.79% FP32 and 93.76% full-integer INT8 accuracy on the eWINE dataset, outperforming three state-of-the-art lightweight baselines: LightMamba (92.03%), TinySA (92.99%), and OCT (91.94%).
  - Deployed the INT8 model on an STM32F401RE Cortex-M4 at 84 MHz, achieving 0.56 ms inference latency with 10.45 KB weights, 3.6 KB RAM, and approximately 10.4K MACs per inference.
  - Reduced weight memory by approximately 3.8× through full-integer INT8 quantization while preserving accuracy within 0.03 percentage points between FP32 and INT8 deployment.
  - **Tech:** UWB, CIR, PCA, TinyML, Python, TensorFlow, C, STM32F401RE, X-CUBE-AI, CMSIS-NN

Publications

- **ATC 2026 – Second-author:**: *Robust Indoor Mobile Robot Localization Using Maximum Correntropy Criterion Unscented Kalman Filter*
- **ATC 2026 – Third-author:**: *Neural Network-Aided Kalman Filtering for Indoor Positioning Autonomous Vehicle System*
- **Journal on Electronics and Communications 2026 – First author:**: *Evaluation of the Applicability of State Estimation Filters for Indoor Robot Positioning*
- **Conference REV-ECIT 2025 – First author:**: *Đánh giá khả năng ứng dụng các bộ lọc ước lượng trạng thái EKF, UKF, PF sử dụng cho định vị robot trong nhà dựa trên công nghệ UWB*

Honors and Awards

- Academic Encouragement Scholarship, Semester 2, 2024–2025 & Semester 1, 2025–2026
- Semi-finalist, Vietnam Electronic Design Competition 2025 - October, 2025
- Encouragement Award, Faculty-level Scientific Research Competition - March, 2025
- Second Prize, Faculty-level Scientific Research Competition - April, 2026